

Lab404: Autonomous Agent Project Plan

Scrum Guide Q&A Agent

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1. Use Case

Problem Statement Scrum Masters and Agile Leaders often complete their certifications but struggle to apply that knowledge in real-world situations. Current alternatives — manual searches or generic LLM answers — don't account for the organization's specific context and practices. This agent bridges that gap by answering questions grounded in both the official Scrum Guide and the organization's own accumulated best practices, which teams can continuously update and which an admin approves before publishing.

Target Users Scrum Masters and Agile Leaders working with Scrum in product and service development contexts — not limited to software teams.

Success Criteria

- Answers are accurate and traceable to the Scrum Guide or a documented best practice
- Advice is practical and contextually relevant, not generic
- Users can add organizational best practices; admin approves before they go live
- Users report reduced time spent searching for answers

Current Process (Manual Today) Users either search the Scrum Guide manually, ask colleagues, or prompt a general-purpose LLM — receiving answers that lack organizational context and may be inconsistent across the team.

2. Technology Stack

Production Stack (MVP V1)

Component	Choice	Justification
Core LLM	OpenAI GPT-4o	Reliable, production-grade, strong reasoning

Component	Choice	Justification
Embeddings	OpenAI text-embedding-3-small	Cost-efficient at scale
Vector DB	Pinecone or Weaviate	Scalable, cloud-hosted, supports multiple collections
Agent Framework	LangGraph	Enables multi-step reasoning and clarifying question flow
RAG Framework	LangChain	Retrieval pipeline over dual knowledge sources
Orchestration	n8n	Notion sync and future Telegram/Slack integration (V2)
Interface	Web app (Streamlit / FastAPI + React)	Accessible, fast to build
Knowledge Base	Scrum Guide PDF + Notion (org best practices)	Dual knowledge sources
Monitoring	LangSmith	LLM call tracing and quality monitoring

Alternatives Considered

- Weaviate vs Pinecone: Both viable; Pinecone simpler to set up for MVP
- LangGraph vs simple LangChain agent: LangGraph chosen for conditional clarifying question flow
- n8n vs custom scheduler: n8n preferred for no-code Notion sync and future integrations

Prototype Stack (Part 2)

Component	Choice
Core LLM	OpenAI GPT-4o
Vector DB	n8n in-memory vector store
Framework	n8n AI Agent (Tools Agent)
Interface	Telegram
Knowledge Base	Scrum Guide PDF only

3. MVP Scope

MVP V1 — Problem-Solution Fit

The goal of V1 is to validate whether users find value in a grounded, source-cited answer over generic LLM responses. Lean and focused.

Included Features:

- Answer questions based on the official Scrum Guide (RAG)
- Answer questions based on org best practices stored in Notion (RAG)
- Ask one clarifying question before answering when context is ambiguous
- Cite sources in every answer (Scrum Guide section or Notion entry)
- Simple web interface
- Admin approval flow for new best practices before they are indexed

Explicitly Out of Scope for V1:

- Conversation memory across sessions
- Telegram / Slack integration
- User authentication
- Analytics dashboard
- Auto-sync of Notion (manual re-indexing by admin)
- Multi-language support
- Integration with Jira or other tools

V2 — If Problem-Solution Fit is Validated

- Conversation memory within session
- Telegram interface
- Users can suggest new best practices via the interface
- Feedback button (👍👎) on answers
- Auto-sync Notion via n8n trigger

V3+

- Slack interface
- Answer history and logs per user
- User authentication
- Analytics dashboard (most asked questions, usage patterns)
- Multi-language support
- Jira integration for context-aware answers
- Support for additional frameworks (SAFe, Kanban)

Success Metrics

Metric	Target
Answer accuracy	>80% validated as correct by SM testers
Source citation rate	100% of answers include a source
User satisfaction	>70% say it's more useful than manual search

Metric	Target
Time to answer	<10 seconds per response
Problem-solution fit	At least 3 SMs say they would use this regularly

4. Risk Assessment

Technical Risks

Risk	Prob.	Imp.	Mitigation
LLM hallucinations — agent answers confidently but incorrectly	MED	HIGH	Always cite source; if no source found, respond with "not covered in available knowledge"
Poor RAG retrieval — wrong chunks returned	MED	HIGH	Test chunking strategy carefully; use metadata filters to separate Scrum Guide vs Notion content
LangGraph complexity — clarifying question flow breaks	LOW	MED	Keep flow simple in V1; limit to one clarifying question max
Notion API changes or downtime	LOW	MED	Cache last known best practices in vector DB;

Risk	Prob.	Imp.	Mitigation
			don't rely on real-time sync in V1

Business Risks

Risk	Prob.	Imp.	Mitigation
Low user adoption — SMs prefer asking colleagues	MED	HIGH	Involve 1-2 SMs in testing early; validate with real questions they've struggled with
Best practices never get added — Notion stays empty	MED	HIGH	Seed Notion with 5-10 best practices before launch; make adding them trivially easy
Admin bottleneck — approval process slows knowledge growth	MED	MED	Keep approval simple (one-click); set 24h SLA expectation

Data Risks

Risk	Prob.	Imp.	Mitigation
Outdated Scrum Guide — new version released	LOW	MED	Pin the version used; display version in UI so users are aware

Risk	Prob.	Imp.	Mitigation
Poor quality best practices submitted	MED	MED	Admin approval acts as quality gate
Notion doc structure changes break the sync	LOW	MED	Use stable Notion page IDs; document expected structure clearly

5. Implementation Plan

Phase 1: Setup and Data Preparation — Sprint 1

Task	Details
Set up development environment	Python, LangChain, LangGraph, Pinecone
Prepare Scrum Guide PDF	Download official PDF, define and test chunking strategy
Set up Notion workspace	Create best practices doc with initial 5-10 entries

Task	Details
Set up vector database	Two collections: Scrum Guide + Notion best practices
Connect Notion API	Read best practices and index into vector DB

Milestone: Both knowledge sources indexed and queryable ✓

Phase 2: Core Agent Development — Sprint 2

Task	Details
Build RAG pipeline	LangChain retriever over both collections
Build LangGraph flow	Build LangGraph flow — Receive question → clarify if needed → retrieve → answer with citations. Clarifying question logic validated via n8n prototype (Tools Agent node).
Prompt engineering	Tune prompts for practical, grounded, cited answers
Admin approval flow	Simple interface to approve/reject submitted best practices

Milestone: Agent answers questions correctly with citations ✓

Phase 3: Interface and Integration — Sprint 3

Task	Details
Build web interface	Streamlit or FastAPI + simple frontend
Connect frontend to agent	API layer between UI and LangGraph
Manual Notion sync	Admin triggers re-indexing when Notion is updated
Internal testing	Test with real SM questions; validate answer quality

Milestone: Working end-to-end application 

Phase 4: Deployment and Monitoring — Sprint 4

Task	Details
Deploy to cloud	Railway or Render (simple and cost-effective)
Set up LangSmith	Trace all LLM calls; monitor answer quality
User acceptance testing	2-3 real Scrum Masters test with real questions

Task	Details
Gather feedback	Document findings; go/no-go decision for V2

Milestone: Live MVP with real users validating problem-solution fit 

Resources Needed

Resource	Details
Team	1 developer (full stack + AI)
LLM costs	OpenAI API — ~\$20-50/month for MVP usage
Vector DB	Pinecone free tier sufficient for MVP
Notion	Free tier sufficient
Hosting	~\$10-20/month (Railway or Render)
Monitoring	LangSmith free tier sufficient for MVP

6. Reflection

Most challenging part of planning: Defining the right MVP scope — the temptation is to plan too much. The lean approach (problem-solution fit first) helped keep V1 focused on the core value proposition: grounded, cited answers combining official Scrum guidance with org-specific knowledge.

What I would do differently: Involve target users (Scrum Masters) earlier in the scoping process to validate assumptions before committing to a technology stack.

Open questions:

- What is the right chunking strategy for the Scrum Guide given its structured format?
- How do we measure "practical relevance" of answers beyond factual accuracy?
- At what point does the Notion best practices base become large enough to justify moving to V2?